

CO1

1. Differentiate between a population and a sample with one example.
 2. Differentiate between a parameter and a statistic.
 3. Estimate the mean of the following observations 12, 15, 18, 20, 25
 4. Estimate the median and mode of the dataset 8, 10, 12, 12, 15, 18, 20
 5. Estimate the range and interquartile range (IQR) for 5, 8, 10, 12, 15, 18, 22, 25
 6. Estimate the variance and standard deviation of 4, 6, 8, 10
 7. Explain any two characteristics of a right-skewed distribution.
 8. Explain the purpose of Exploratory Data Analysis (EDA).
 1. The following dataset represents the ages (in years) of employees: 24, 26, 28, 30, 32, 34, 36. Estimate the first quartile (Q_1) and third quartile (Q_3).
 2. The marks of 10 students are: 42, 48, 50, 55, 58, 60, 62, 65, 68, 72. Estimate: Mean, Median, Mode
 3. The daily temperatures ($^{\circ}\text{C}$) recorded over a week are: 28, 30, 29, 31, 35, 34, 33. Estimate: Range, Variance, Standard Deviation
 4. The following summary statistics are obtained for monthly electricity consumption (kWh): Minimum = 180, Q_1 = 240, Median = 290, Q_3 = 360, Maximum = 620. Estimate: Interquartile Range (IQR), Lower Fence, Upper Fence. Interpret whether 620 kWh is an outlier.
 5. Explain the importance of IQR in identifying outliers.
 6. Differentiate between a Histogram and a Box Plot.
 7. Explain the basic steps involved in Exploratory Data Analysis (EDA)
 8. A data scientist records the daily sales (in units) of a product for 10 days: 45, 50, 52, 48, 55, 60, 62, 58, 50, 70
 - a) Calculate the mean, median, and mode.
 - b) Determine the range and standard deviation.
9. A company has 5,000 employees. A researcher selects 100 employees and records their monthly salaries.
- a) Illustrate the difference between population and sample in this context.
 - b) Illustrate the difference between a parameter and a statistic.
 - c) The salaries (₹ in thousands) of 10 sampled employees are 25, 28, 30, 35, 40, 42, 45, 50, 55, 60. Calculate the mean salary and variance.
10. The response times (in milliseconds) of a web server are 120, 130, 125, 140, 135, 128, 500
- a) Calculate the mean and median.
 - b) Which measure of central tendency is more appropriate for this dataset? Justify your answer.
 - c) Calculate the range and interquartile range (IQR).
11. A dataset contains the following observations: 12, 15, 18, 20, 22, 25, 28, 30, 35
- a) Compute the mean, median, and mode.
 - b) Compute the variance and standard deviation.
 - c) Interpret the variability of the dataset based on the standard deviation.
12. A data analyst collected the following customer purchase amounts (₹): 120, 135, 140, 145, 150, 152, 155, 158, 160, 165, 170, 175, 180, 190, 300. Perform an exploratory data analysis by

- a) Finding the five-number summary (Minimum, Q1, Median, Q3, Maximum).
- b) Calculate the IQR and determine whether the dataset contains any outliers using the $1.5 \times \text{IQR}$ rule.
- c) Based on your findings, comment on the likely shape of the distribution.

13. The following summary statistics are obtained for the monthly electricity consumption (kWh) of households: Minimum = 180, Q1 = 240, Median = 290, Q3 = 360, Maximum = 620.

- a) Calculate the IQR.
- b) Determine the lower and upper fences for outlier detection.
- c) Determine whether the maximum value is an outlier.
- d) Sketch the corresponding box plot and comment on the distribution.

14. The following frequency distribution represents the ages of customers visiting a retail store.

Age (Years)	Frequency
20–30	4
30–40	8
40–50	15
50–60	10
60–70	6
70–80	2

- a) Draw a histogram for the given data.
- b) Determine whether the distribution is approximately symmetric, positively skewed, or negatively skewed.
- c) Determine whether the mean or median is likely to be larger and justify your answer.

15. A dataset has the following descriptive statistics: Mean = 68, Median = 61, Standard Deviation = 14, Pearson's coefficient of skewness: $Sk = \frac{3(\text{Mean} - \text{Median})}{\text{Standard Deviation}}$. Also, the kurtosis coefficient is 4.8.

- a) Calculate the coefficient of skewness.
- b) Determine the distribution as positively skewed, negatively skewed, or symmetric.
- c) Determine whether the distribution is platykurtic, mesokurtic, or leptokurtic.

16. A data scientist records the following response times (ms): 82, 85, 86, 88, 90, 91, 92, 93, 95, 96, 98, 100, 102, 150

- a) Compute the five-number summary.
- b) Determine whether a histogram or a kernel density estimate (KDE) would better reveal the distribution pattern, considering the data.
- c) Determine whether the dataset is likely to be right-skewed, left-skewed, or symmetric based on the given values.

17. The marks obtained by 50 students are follows bellow and Calculate: Mean, Median, Mode. Interpret the distribution.

Marks	Frequency
0–10	5
10–20	8
20–30	12
30–40	15
40–50	10

18. The following summary statistics are obtained for the monthly electricity consumption (kWh) of a sample of households: Minimum = 180, $Q_1 = 240$, Median = 290, $Q_3 = 360$, Maximum = 620

Answer the following:

- Calculate the Interquartile Range (IQR).
- Determine the Lower Fence and Upper Fence for detecting outliers.
- Determine whether 620 kWh is an outlier.
- Draw the corresponding Box Plot and comment on the distribution.

19. A dataset has the following five-number summary: Minimum = 12, $Q_1 = 20$, Median = 28, $Q_3 = 36$, Maximum = 60. Construct a Box Plot, determine whether outliers exist, and comment on the distribution.

20. Illustrate Mean, Median and Mode with respect to symmetric data, skewed data and data containing outliers. Support your answer with examples.

CO2

1 A machine learning classifier correctly predicts a class with probability 0.92. Model one prediction as a Bernoulli random variable. Estimate its mean.

2 A quality inspector randomly checks 6 electronic components. The probability that a component is defective is 0.1. Estimate the probability that exactly one component is defective.

3 A chatbot successfully answers a customer's query with probability 0.8 on each attempt. Estimate the probability that the first successful response occurs on the third attempt.

4 A web server receives an average of 4 requests per minute. Estimate the probability of receiving exactly two requests in one minute.

5 A random variable follows a Uniform distribution over $[5,15]$. Estimate its expected value.

6 The heights of students are normally distributed with mean 170 cm and standard deviation 8 cm. Estimate the Z-score corresponding to a height of 182 cm.

7 The lifetime of an IoT sensor follows an exponential distribution with rate parameter $\lambda = 0.5$ failures/year. Estimate the mean lifetime of the sensor.

8 A Gamma distribution has shape parameter $\alpha = 4$ and scale parameter $\theta = 2$. Find its expected value.

9 Explain the Central Limit Theorem.

10 Explain Maximum Likelihood Estimation (MLE).

1 A fraud detection system identifies a fraudulent transaction with probability 0.96. Model the outcome as a Bernoulli random variable. Estimate: a) Mean b) Variance

2 A machine learning model correctly classifies an image with probability 0.85. If 8 images are selected independently, estimate the probability that exactly 6 images are classified correctly.

3 A recommendation engine recommends the correct product with probability 0.30 on each attempt. Estimate:

- a) Probability that the first success occurs on the 4th recommendation
- b) Expected number of recommendations until the first success.

4 A cloud server receives an average of 5 login requests per minute. Estimate the probability that during one minute: a) Exactly 3 requests arrive. b) No request arrives.

5 The response time (seconds) of a web application is uniformly distributed between 2 s and 8 s. Estimate: a) Mean b) Variance

6 The IQ scores of applicants are normally distributed with Mean = 100 and Standard deviation = 15. Estimate the probability that an applicant scores between 85 and 115. (Use the Standard Normal Table.)

7 The lifetime (years) of a wireless sensor follows an exponential distribution with $\lambda = 0.25$. Estimate the probability that the sensor lasts more than 5 years.

8 The average battery life of smartphones is 12 hours with population standard deviation 2 hours. A random sample of 64 phones is selected. Estimate the standard error of the sample mean.

9 Differentiate between the following distributions by giving one suitable Data Science application for each: Binomial Distribution, Poisson Distribution

10 Explain the importance of the covariance matrix in the Multivariate Normal Distribution.

1 An email spam filter classifies an email correctly with probability 0.93. Model a single classification as a Bernoulli random variable. Calculate: a) Mean b) Variance c) Standard deviation

2 A disease detection model correctly identifies infected patients with probability 0.80. If 12 patients are tested independently, find the probability that:

- a) Exactly 9 patients are correctly identified.
- b) At least 10 patients are correctly identified.

2.a) The annual growth factors of a company's revenue over six consecutive years are given below:

Year Growth Factor

1	1.08
2	1.12
3	0.96
4	1.15
5	1.10
6	1.05

1. Calculate the Geometric Mean (GM) of the growth factors. (5 Marks)
2. Interpret the result by stating the average annual growth rate (%) of the company's revenue over the six-year period.

3 An online advertisement is clicked with probability 0.15. Find:

- a) The probability that the first click occurs on the fifth impression.
- b) The expected number of impressions before the first click.

4 A customer support center receives an average of 8 support tickets per hour. Find the probability that in one hour:

- a) Exactly 6 tickets are received.
- b) More than 8 tickets are received.

5 The processing time (milliseconds) of an AI inference engine follows a Normal distribution with: Mean = 120 ms and Standard Deviation = 15 ms. Find: a) $P(X < 135)$ b) $P(110 < X < 140)$

(Use the Standard Normal Table.)

6 The daily number of transactions processed by a payment gateway has: Population Mean = 450 and Population Standard Deviation = 60. A random sample of 36 days is selected. Using the Central Limit Theorem, calculate:

- a) The standard error of the sample mean.
- b) The probability that the sample mean exceeds 470 transactions.

7 The average delivery time of an e-commerce platform is 4.5 days with a population standard deviation of 1.2 days. A random sample of 64 orders is selected. Find:

- a) The standard error.
- b) The probability that the sample mean delivery time is less than 4.8 days.

8 A Bernoulli random variable produced the following observations: 1, 0, 1, 1, 0, 1, 1, 1, 0, 1. Using the Maximum Likelihood Estimation (MLE) method, estimate the parameter p .

9 Demonstrate the Central Limit Theorem with a suitable Data Science example. Illustrate why it is important in statistical inference.

10 Demonstrate the Maximum Likelihood Estimation (MLE) procedure. Illustrate the steps involved in estimating model parameters from observed data.

1 A machine learning classifier correctly predicts an image category with probability 0.88. If 15 images are classified independently, calculate:

- a) Probability that exactly 12 images are correctly classified.
- b) Probability that at least 13 images are correctly classified.
- c) Mean and variance of the distribution.

2 A cloud server receives an average of 12 API requests per minute. Using the Poisson distribution, determine:

- a) Probability of receiving exactly 10 requests.
- b) Probability of receiving at least 15 requests.
- c) Expected number and variance of requests.

3 The scores of students in a Data Science aptitude test are normally distributed with: Mean = 75 and Standard Deviation = 12. Find:

- a) $P(X > 90)$
- b) $P(65 < X < 85)$
- c) The percentage of students scoring below 60.

Use the Standard Normal Table.

4 The lifetime (hours) of an electronic component follows a Uniform Distribution over the interval (100, 160). Calculate: a) Mean b) Variance. If the lifetime instead follows a Gamma Distribution with shape parameter $\alpha = 5$ and scale parameter $\theta = 20$, determine: c) Mean d) Variance. Illustrate the differences between the two models.

5 The daily number of online purchases on an e-commerce platform has: Population Mean = 850 and Population Standard Deviation = 120. A random sample of 64 days is selected. Using the Central Limit Theorem, determine:

- a) Standard error of the sample mean.
- b) Probability that the sample mean exceeds 875 purchases.
- c) Comment on the applicability of the CLT.

6 The average response time of a web server is 2.8 seconds with a population standard deviation of 0.6 seconds. A random sample of 49 requests is selected. Calculate:

- a) Standard error.
- b) Probability that the sample mean response time is between 2.6 and 3.0 seconds.

7 A Bernoulli random variable generated the following observations: 1, 1, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 1, 1, 0. Using the Maximum Likelihood Estimation (MLE) approach:

- a) Estimate the parameter p .
- b) Write the likelihood function.
- c) Write the log-likelihood function.

8 A two-dimensional random vector (X,Y) has the following parameters: Mean vector: $\mu = \begin{bmatrix} 70 \\ 75 \end{bmatrix}$
Covariance matrix: $\Sigma = \begin{bmatrix} 25 & 10 \\ 10 & 16 \end{bmatrix}$. Answer the following:

- a) Interpret the variances of X and Y .
- b) Determine the covariance between X and Y .
- c) Compute the correlation coefficient between X and Y .
- d) Comment on the relationship between the two variables.

9 Demonstrate the Central Limit Theorem. Illustrate its significance in Data Science, particularly in model evaluation, confidence intervals and hypothesis testing.

10 Demonstrate the concept of Maximum Likelihood Estimation (MLE). Illustrate its role in estimating parameters for Bernoulli, Binomial and Normal distributions used in machine learning.